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The yield of sugar beet in relation to weather and length of growing season

By R. K. SCOTT, S. D. ENGLISH, D. W. WOOD and M. H. UNSWORTH
University of Nottingham, School of Agriculture, Sutton Bonington, Nr. Loughborough, Leicestershire

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SUMMARY

An experiment at Sutton Bonington in 1971 tested the effects of sowing and harvesting dates on yield of sugar beet. Seed sown on 28 March germinated slowly in the cold soil and crop establishment was poor. From early April until the beginning of May the loss in yield of sugar consequent upon delaying sowing by 1 week was 0·4 t/ha. For later sowings the yield loss was greater. During the harvest period the rate of increase in sugar yield was the same for all sowings. Yields of crops sown and harvested on different dates were closely correlated with the amount of solar radiation intercepted by the leaf canopy.

INTRODUCTION

Sowing-date experiments with sugar beet recently reported by Hull & Webb (1970) confirmed the results of earlier experiments (Anon, 1960; Holmes & Adams, 1966) that delay in sowing after early April caused progressively greater yield loss. March sowings outyielded early April sowings in 4 years out of 5 and it seems possible that if varieties are produced which are even less susceptible to bolting, and perhaps better able to grow at low temperatures than present varieties, still earlier sowing than is now normal would be beneficial. With potatoes the effectiveness of early planting is limited by early senescence, so that later plantings may yield as well if not better provided the growing season is not curtailed (Bremner & Radley, 1966). No such tendency exists in sugar beet which is biennial and continues growing as long as the environment allows. Experiments of Holmes & Adams (1966) in Lincolnshire indicated that early-sown beet increased in yield more during autumn than late-sown beet. In Hull & Webb's (1970) experiments sugar yield increased with length of growing period in all years but the relationship was non-linear because increments in yield diminished as the growing period was prolonged either by earlier sowing or later harvest.

The experiment reported here tested how changes in soil heat content affected the establishment of crops sown on different dates and how sowing date affected the rate of sugar accumulation in autumn. Relationships between sugar yield, length of growing season and environmental factors were investigated.

EXPERIMENTAL PROCEDURES

The effect of temperature on germination was determined in constant-temperature rooms operating at 3·5, 5, 7, 10 and 15 °C. Four replicates of 25 clusters were placed on damp filter papers in Petri dishes in polyethylene bags. Numbers germinating were counted daily.

In the field, soil temperature at the level of seeds, 20–25 mm below the surface, was recorded with a mercury-in-steel thermograph. Average temperatures during 24 h periods were calculated from readings at 2 h intervals from the charts.

The field experiment was made on an imperfectly drained sandy loam of the Astley Hall series, overlying Keuper Marl. Previous cropping was potatoes (1968), winter wheat (1969) and spring barley (1970). After ploughing in autumn 1970, 0·62 t/ha of kainit was applied to the experimental site and 124 kg N/ha, 62 kg P_2O_5/ha and 62 kg K_2O/ha were applied in spring before the first sowing. The first sowing was made on 28 March as soon as the soil was dry enough to prepare a seed bed. Each main plot was cultivated separately just before sowing to prepare a fine, firm seed bed. The experiment was laid out in four randomized blocks each containing five main plots for the sowing date treatments (28 March, 8 April, 22 April, 5 May, 19 May). The main plots were split into five subplots for the harvesting date treatments (22 September, 6 October, 20 October, 19 November, 1 December). Main plots had ten rows, 0·51 m apart and 19·2 m long; a subplot consisted of 9·7 m^2 from the centre eight rows. Seed of variety Sharpe's Klein E, rubbed and graded to 3·2–4·4 mm, was

precision-drilled at 30 mm spacing and 20 mm deep. Seedlings were counted daily. Plants were singled by hand to a target population of 75000/ha. Weeds were controlled by steerage and hand-hoeing rather than herbicides, which might have differed in their efficacy on different sowings. All plots were sprayed with demeton-s-methyl on 20 July to control aphids and few plants had virus yellows.

Samples of plants were taken from the first and fourth sowings, from an area of 3·1 m^2 per plot, at singling (16 May and 1 June respectively), on 9 July, 9 August, 6 September and at each harvest date. The plants were separated into leaf, stem and petiole (including the crown) and root. Leaf-area indices were estimated by the leaf-disk method of Watson & Watson (1953) as modified by Bremner & Taha (1966).

Sunshine hours, total solar radiation, maximum and minimum air temperatures and soil-heat flux were recorded at an Agrometeorological Station 1 km from the experimental site. The soil-heat flux (W/m^2), a measure of the rate of net heat flow into or out of the soil, was determined with flux plates (Slatyer & McIlroy, 1961) at a depth of 70 mm under a grass surface at 15 min intervals.

In seeking factors which may limit early crop development, direct measurements of soil temperature at a single point are difficult to interpret, first because they show rapid fluctuations which make trends difficult to discern and secondly because the temperature in the drill does not necessarily represent temperatures throughout the rooting profile. The first objection may be removed by calculating cumulative soil temperature above or below some arbitrary base-line which we have taken as 5 °C. The cumulative soil temperature was obtained by accumulating the difference between this base temperature and the daily mean averaged from the readings at 2 h intervals on the thermograph. An alternative approach which shows the heat stored in the soil is the calculation of 'soil-heat content', which we define as the product (strictly the integral) of soil-heat flux and time (J/m^2). This represents the net energy transferred through unit area of soil at the level of measurement. This energy changes soil temperature at a rate determined by the thermal conductivity and specific heat of the soil. Cumulative soil-heat content was calculated during March, April and May as an indication of the pattern of soil heating. In the later stages of plant growth, plant development is often better correlated with air rather than soil temperatures (van Wijk, 1965) and the day degree has frequently been used for this purpose (Milbourn & Hardwick, 1968). Day degrees were cal-

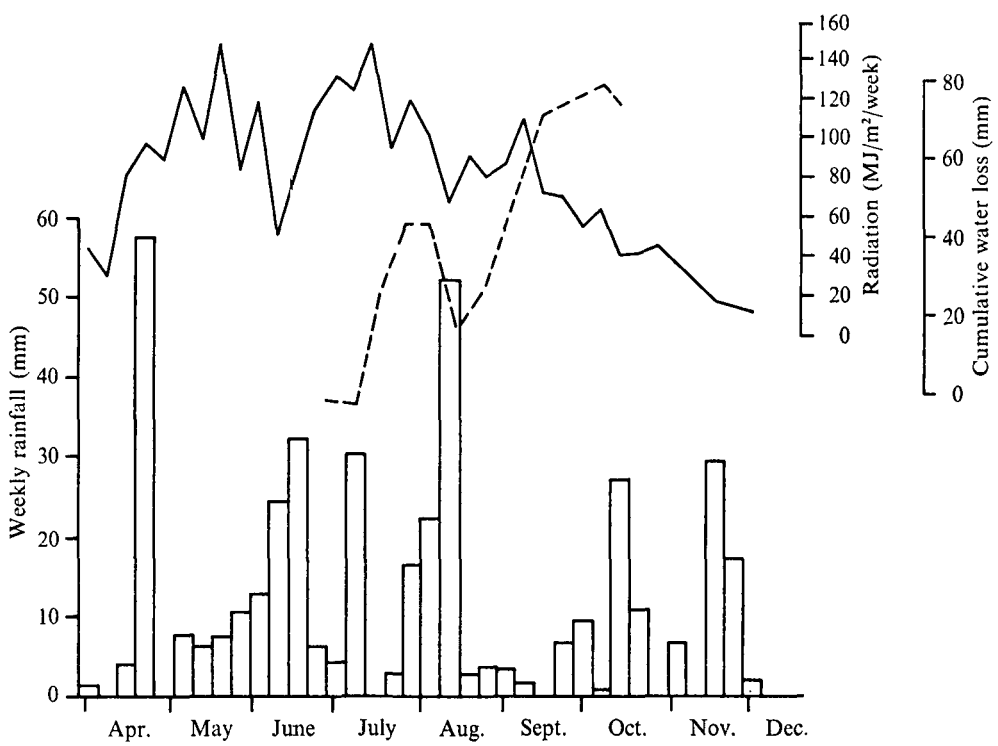

Fig. 1. Meteorological data. ——, Weekly radiation; ---, cumulative water loss; □, rainfall.

culated from maximum and minimum air temperatures using Meteorological Office tables and assuming a base temperature of 5 °C.

RESULTS

General

During 1971, none of the sowings appeared severely stressed by lack of water at any time. Rainfall in April was above average due to heavy rain on 23 April, shortly after the third sowing and just before emergence of the first and second sowings. Frequent rain in June maintained the soil at field capacity and the soil profile did not dry appreciably until early July (Watkins, 1972); the cumulative net water loss from the top 1·6 m of the soil reached 79 mm in early October (Fig. 1). In the same year, Hull (1972) found no response to irrigation at Broom's Barn Experimental Station. Figure 1 also shows weekly solar radiation throughout the season, indicating the rapid increase in radiation receipt during April and May and the reduction during June. Soil temperature (Fig. 2) increased only slowly until mid-April and fluctuated considerably during unsettled weather at the end of the month. Trends in soil and air temperatures, soil-heat content and radiation are seen more easily in the cumulative curves of Fig. 2.

Effect of temperature on germination

In the laboratory few seeds germinated, and only very slowly, below 5 °C (Fig. 3*a*). Thus the choice of 5 °C as the base temperature for calculation of cumulatve soil temperature and day degrees is justified.

Effects of sowing date

Between the first and second sowings, average soil temperature was 6·5 °C (Fig. 2) and germination proceeded slowly (Fig. 3*b*). The soil started to warm and the temperature between sowing and emergence of the second sowing averaged 9·1 °C. Emergence of the first sowing was only marginally ahead of the second. Table 1 shows that the delay affected the vigour of the earliest sowing and it was difficult to obtain uniformly spaced plants.

Temperatures increased and the interval between sowing and emergence shortened with later sowing, establishment improved and desired plant populations were readily obtained. The fifth sowing was slightly slower to emerge than the fourth as the temperature at the end of May was 2 °C lower on average (Fig. 2).

The first sowing emerged 3 weeks before the fourth but attained a leaf area index (LAI) of 3·5, the optimum for sugar beet (Goodman, 1966;

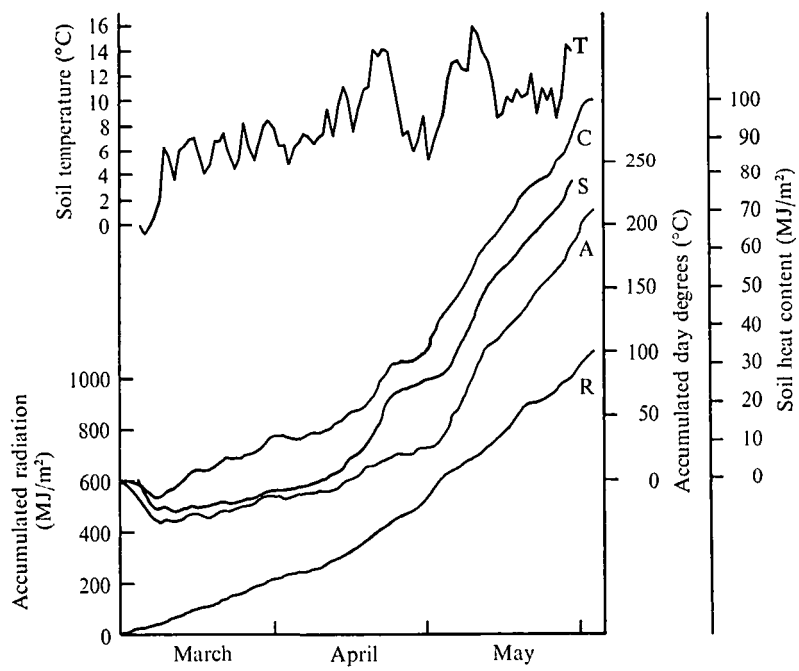

Fig. 2. Changes in soil temperature (T), accumulated day degrees for air (A) and soil (S) temperatures, soil heat content (C) and solar radiation (R) during spring 1971.

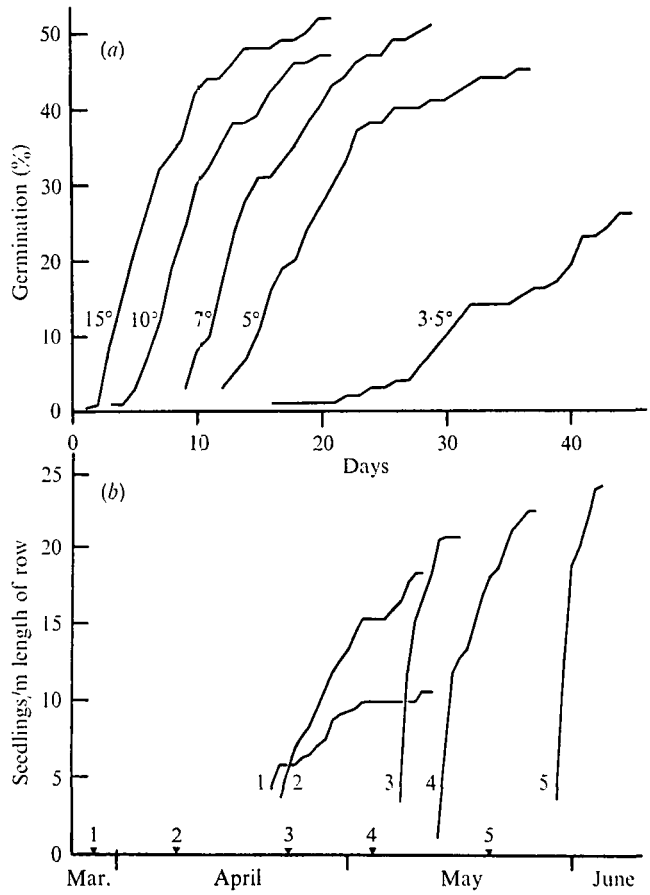

Fig. 3. Effects of (*a*) a range of constant temperatures on germination and (*b*) sowing date on emergence in the field of Sharpe's Klein E. 1, sown 28 March; 2, sown 8 April; 3, sown 22 April; 4, sown 5 May; 5, sown 19 May.

Bremner, El Saeed & Scott, 1967) only 10 days earlier (Fig. 4). By early August LAI's were similar and during September and October the LAI of the fourth sowing exceeded that of the first sowing because individual leaves on late sown plants were larger and thinner, although fewer. Watson & Baptiste (1938) reported a similar but more pronounced effect. It seems unlikely that this slightly greater leaf area in autumn compensated for the substantially smaller (36%) leaf-area duration (LAD) during the long days in late May, June and July (Fig. 1).

The first sowing had greater dry weight than the fourth throughout the season. As the season progressed the storage root constituted an increasing proportion of total dry weight, consistently 2–3% more of the total dry weight of early than late sown plants at each harvest. This simply reflected the normal pattern of change in dry-matter distribution with increasing dry weight, since the ratio (root dry weight)/(total dry weight) was not affected by sowing date.

Sugar yields of crops sown on 28 March and 8 April were similar at each harvest (Fig. 5). Delay in sowing between early April and early May resulted in the loss of 0·4 t sugar/ha/week with a greater penalty for further delay. The overall yield advantage from sowing in early April rather than 6 weeks later in mid-May was similar to the gain from delaying harvest by 8–9 weeks from late September until early December.

Effects of harvest date

The autumn was warmer and drier than usual but mid-October was cold and wet (Table 2). This period caused a marked discontinuity in the pattern of increase in sugar yield. Average rates of increase were, from 22 September to 6 October, 0·8 t/ha/

Table 1. *Effects of sowing date*

Date of sowing	No. of days to 50% emergence	Final establishment plants/m of row	Final population ('000/ha)	Sugar yield average of all harvest dates (t/ha)
28 March	27	11	72·3	10·53
8 April	19	19	76·2	10·56
22 April	15	21	76·0	9·48
5 May	10	23	75·0	8·80
19 May	12	25	79·4	7·00
	±0·9	±1·4	±1·34	±0·155

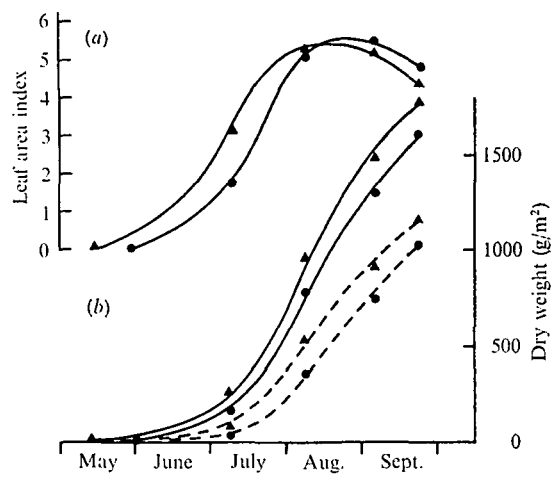

Fig. 4. (*a*) ——, Leaf-area development. (*b*) Root (– –) and total (—) dry-matter accumulation, for sowing on 28 March (▲) and 5 May (●) in 1971.

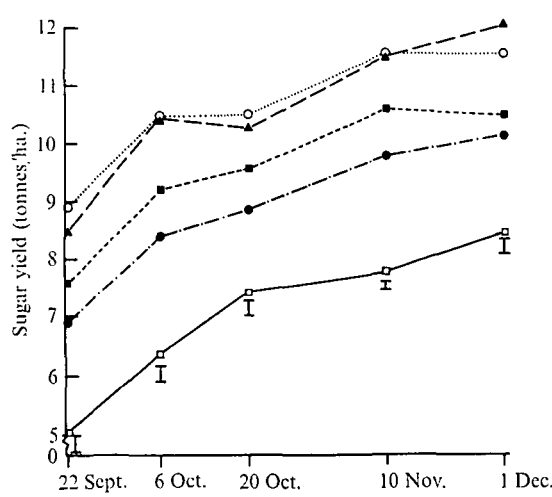

Fig. 5. Accumulation of sugar in the autumn by crops sown on 28 March (▲– –▲), 8 April (○· · · ·○), 22 April (■– – –■), 5 May (●—·—●) and 19 May (□——□). Bars denote 1 S.E. of a point.

Table 2. *Relation between autumn weather and rates of increase in root yield*

Year	Site	Period of investigation	Sunshine (h)	Accumulated day-degrees	Change in root yield (t/ha)
1963	Broom's Barn	23 Sept. to 14 Oct.	85	252	5·4
		15 Oct. to 12 Nov.	81	310	5·4
1964	Broom's Barn	22 Sept. to 19 Oct.	157	268	5·2
		20 Oct. to 16 Nov.	85	89	0·7
1965	Broom's Barn	21 Sept. to 18 Oct.	132	340	7·6
		19 Oct. to 15 Nov.	124	95	0·3
1971	Sutton Bonington	23 Sept. to 6 Oct.	47	189	5·9
		7 Oct. to 20 Oct.	37	38	1·7
		21 Oct. to 4 Nov.	69	131	2·0

week, from 7 to 20 October when the weather was cold and wet, 0·18 t/ha/week, and from 21 October to 10 Novemer, 0·31 t/ha/week. For later sowings, the check to growth seemed progressively less pronounced (Fig. 5). Throughout the autumn, however, the overall course of increase in sugar yield was essentially parallel for all sowings. The first and last sowings increased in yield between 22 September and 1 December by strikingly similar amounts – 3·56 and 3·41 t/ha, respectively.

Relationship between sugar yield and growing season

The growing periods resulting from different sowing and harvesting dates ranged from 126 to 247 days. The relationship between growing period and sugar yield shown in Fig. 6 and Table 3, equation 1, was similar to that obtained by Hull & Webb (1970). The non-linear relationship between growing period and yield suggests that environmental factors may have restricted growth early and late in the season.

Since the earliest sowings were slow to emerge, sugar yield might have been more closely related to the interval between emergence and harvest than between sowing and harvest. This proved not so; the relationship was again non-linear and a quadratic function accounted for only 83% of the variance (equation 2). In a year with adequate rainfall when sugar beet did not respond to irrigation (Hull, 1972), factors likely to determine yield potential are air temperature and hours of sunshine (or solar radiation). The relationship between sugar yield and accumulated day-degrees from sowing until harvest was linear and accounted for 88% of the variance (equation 3). This clearly reflects the predominant effect of temperature early and late in the season.

Hours of bright sunshine are more often recorded than radiation, at weather stations. Figs. 2 and 3 suggest that soil temperature early in the season was more affected by rain and by air temperature than by radiation, hence unless a species was particularly light-sensitive in germination the direct influence of hours of bright sunshine would have been confined largely to the period between emergence and harvest. This relationship was, however, non-linear and a quadratic function accounted for 90% of the variation in yield (equation 4). In contrast, the total radiation incident between emergence and harvest was linearly related and well correlated with yield, accounting for 94% of the variation. Growth analysis data for the first and fourth sowings allowed us to test whether the relationship could be improved by considering the radiation intercepted by the leaf canopy calculated from Beer's Law:

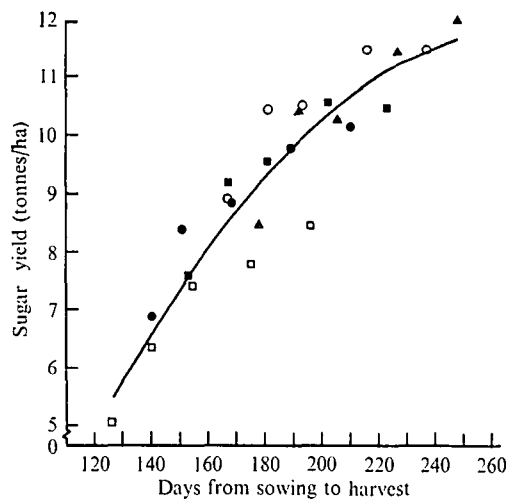

Fig. 6. The effect of length of growing season on sugar yield of crops sown on 28 March (▲), 8 April (○), 22 April (■), 5 May (●) and 19 May (□). ——, Average regression line calculated by equation 1 in Table 3.

$$I_L = I_0 e^{-kL},$$

Table 3. *Regression equations of sugar yield on length of growing season and meteorological parameters*

Eqn no.		Variation in yield accounted for (%)
1.	$Y = -9{\cdot}89(\pm 4{\cdot}13) + 0{\cdot}16(\pm 0{\cdot}04)\,A - 0{\cdot}28 \times 10^{-3}\,(\pm 0{\cdot}12 \times 10^{-3})\,A^2$. Y = sugar yield (t/ha), A = no. of days between sowing and harvest.	88
2.	$Y = -11{\cdot}98(\pm 5{\cdot}03) + 0{\cdot}20(\pm 0{\cdot}06)\,B - 0{\cdot}43 \times 10^{-3}\,(\pm 0{\cdot}18 \times 10^{-3})\,B^2$. B = no. of days between emergence of 50 % of the seedlings and harvest.	83
3.	$Y = -9{\cdot}46(\pm 1{\cdot}36) + 7{\cdot}6 \times 10^{-3}\,(\pm 0{\cdot}6 \times 10^{-3})\,C$. C = no. of day-degrees accumulated between sowing and harvest.	88
4.	$Y = -11{\cdot}67(\pm 3{\cdot}99) + 0{\cdot}035(\pm 0{\cdot}010)\,D - 1{\cdot}2 \times 10^{-5}\,(\pm 0{\cdot}6 \times 10^{-5})\,D^2$. D = no. of hours of sunshine between emergence of 50 % of the seedlings and harvest.	90
5.	$Y = -5{\cdot}25(\pm 0{\cdot}72) + 0{\cdot}25(\pm 0{\cdot}01)\,E$. E = Quantity of radiation (MJ/m²) received between emergence of 50 % of the seedlings and harvest.	94
6.	$Y = -5{\cdot}64(\pm 1{\cdot}62) + 6{\cdot}9 \times 10^{-3}\,(\pm 0{\cdot}7 \times 10^{-3})\,F$. F = Quantity of radiation (MJ/m²) received between emergence of 50 % of the seedlings and harvest (first and fourth sowings only).	92
7.	$Y = -2{\cdot}68(\pm 1{\cdot}07) + 8{\cdot}9 \times 10^{-3}\,(\pm 0{\cdot}8 \times 10^{-3})\,G$. G = Quantity of radiation (MJ/m²) intercepted by the crop (first and fourth sowings only).	95

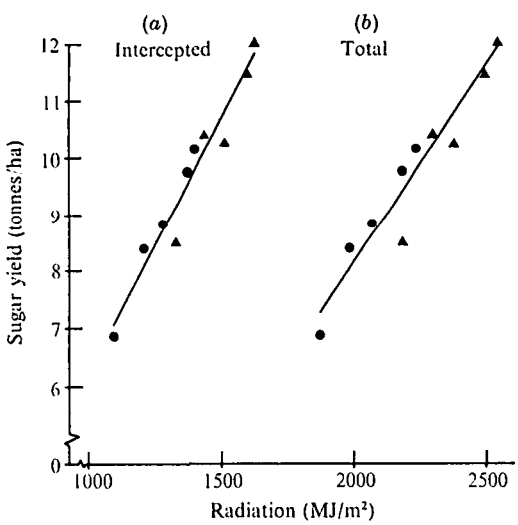

Fig. 7. The relationship between sugar yield and radiation, (*a*) intercepted (equation 7), and (*b*) total (equation 6), for sowing on (▲) 28 March and (●) 5 May. ——, Average regression lines based on equations 6 and 7 respectively in Table 3.

where I_L = radiation below a canopy with leaf area index L, I_0 = radiation incident at the top of the canopy, k = extinction coefficient, and assuming k = 0·53 (Bremner *et al.* 1967).

For the first and fourth sowings the regression lines accounted for 92% of the variation in yield when all radiation incident between emergence and harvest was considered and 95% when only the proportion intercepted was included (Fig. 7).

DISCUSSION

Germination and emergence

The earliest sowing in 1971 was slow to germinate and produced a sparse crop (Fig. 3, Table 1). Hull & Webb (1970) did not experience this difficulty and Holmes & Adams (1966) report sparser stands from early sowing in only one in ten experiments. As more of the crop is sown with monogerm seed, sometimes drilled to a stand, improved methods of seed and seedling protection will be needed if full advantage is to be taken of the potential for a longer growing season offered by early sowing.

Weather and yield

In 1971 cold wet weather in October decreased the rate of sugar accumulation (Table 2). Cold weather between mid-October and mid-November at Broom's Barn in 1964 and 1965 similarly restricted the rate (Table 2). Presumably the relationship between sugar yield and growing period (Table 3, equation 1) is non-linear because growth,

especially of the earliest sowings, is restricted by cold in early spring, and by low temperatures and irradiance in autumn. In summer the growth of late sowings may be restricted, not by the potential of the environment, but by the crop having inadequate leaf area to take advantage of sunlight during the long mid-summer days (Figs. 1, 4).

The well-correlated linear relationships between yield and day-degrees, yield and incident radiation, and yield and intercepted radiation (equations 3, 5, 6 and 7) show the importance of and interrelation between radiation and temperature in determining yield. The relationship between sunshine hours and yield was non-linear, probably because low temperature restricts the effective utilization of bright sunshine early and late in the season. Also, hours of bright sunshine early and late in the season when the sun is less intense 'contain' less radiation.

Since temperature appears to have such an important effect on early development of sugar beet it would be advantageous to choose the optimum sowing date on the basis of a temperature that can be measured easily. In 1971, the latest sowing date which did not restrict yields of sugar was 8 April; since many seeds sown earlier failed to produce seedlings, it was a better time for sowing than earlier. Soon after this, soil temperatures increased rapidly (Fig. 2, cumulative curve). At the same time there was a rapid increase in cumulative soil heat content, and, to a lesser extent, day-degrees. In the absence of soil temperature or heat flux measurements, cumulative day-degrees would have given a useful measure of the suitability of soil conditions for crop growth. From Broom's Barn records it is possible to test whether in other experiments the optimum sowing date also corresponds with the time of rapid increase in accumulated day-degrees. In 4 out of 6 years (1963–7 and 1971) for which records are available this was clearly so (Fig. 8). In 1963 and 1965 the first sowing coincided with the increase in day-degrees and yields were greater than from later sowings. In 1964 and 1967 the increase occurred about the time of the second sowing, which produced as much sugar as the earlier-sown crop. There are clearly limits to the use of day-degrees to predict soil condition since the initial temperature of the soil in spring depends partly on the severity of the winter, and the amount of heat required to reach a threshold temperature will vary from year to year. Thus in 1966 and 1971 at Broom's Barn the first sowing gave the largest yields, yet the increase in day-degrees did not occur until the time of the second sowing. Field records indicate that soil water was not limiting in either year and as March was not severely cold the soil may have been sufficiently warm at the time of the first sowing; in 1966, February air temperatures were above

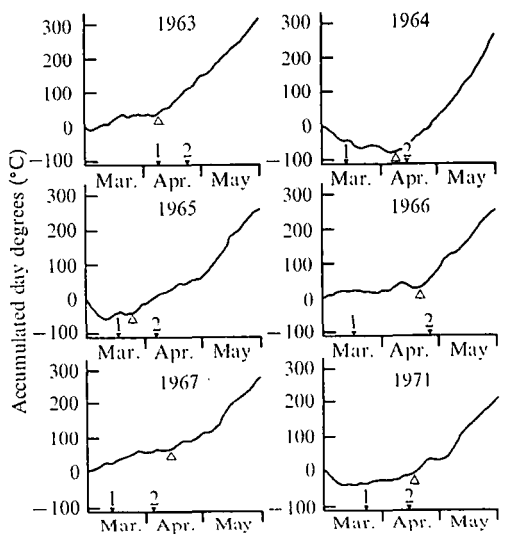

Fig. 8. Changes in accumulated day-degrees and the onset of the rapid increase in accumulation (△) in relation to the first two sowings (▼) at Broom's Barn in 1963–7 and 1971.

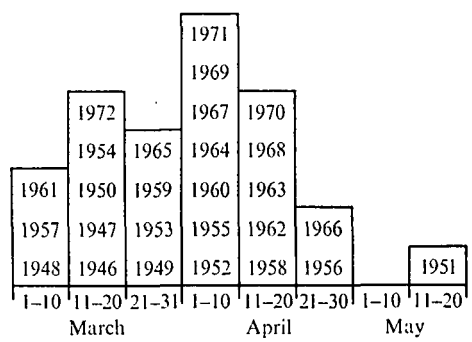

Fig. 9. Date of onset of rapid increase in accumulated day-degrees at Sutton Bonington 1946–72.

average and the winter of 1970/1 was generally mild.

The accumulated day-degrees at Sutton Bonington were calculated for the period 1946–72. The basic pattern of increase was similar in each year, but the onset of the rapid rate of increase varied from the beginning of March to the end of April (Fig. 9); in some years the date was difficult to determine precisely. It would be advantageous to be able to define this point objectively. However, simple mathematical analysis suggests that at Sutton Bonington the shape of the annual temperature wave varies in a complex manner.

Fig. 9 shows that in 12 of the last 27 years accumulated day-degrees have increased rapidly

in March, so that March sowings of Sharpe's Klein E would have been beneficial to yield at Sutton Bonington. There is a tendency, however, for springs to be later and this advantage would only have occurred in 7 of the last 20 years.

The results of this experiment emphasize that early leaf development is necessary to get large yields. The yield advantage from early sowing seemed to stem from the crop making greater use of incident radiation. The earliest emerging crop intercepted 62% of radiation incident between emergence of this crop and harvest; the fourth sowing intercepted 54% over the same period. Scott & Bremner (1966) demonstrated that raising seedlings under glass and transplanting them into the field at normal sowing time gave a crop that intercepted more radiation and yielded more than one sown directly in the field. Alternative ways of hastening early leaf development are to breed varieties capable of germination and leaf expansion at low temperatures or to select for extreme bolting resistance and sow in autumn. Such a crop would over-winter in the seedling stage and start growth with its foliage already established as soon as temperatures rose in spring. Both approaches are currently being investigated.

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